PK-PD Practice 1

Module 7b - PK-PD Relationship · Daily practice sheet · No date

Name _____ Date _____

A certain antibiotic that follows a one-compartment open model has an apparent volume of distribution of approximately 15 L and a first-order elimination half-life of approximately 5 hours and is characterized by a minimum effective concentration of 10 mcg/mL.

a What is the teff for a single 500-mg IV bolus dose of this antibiotic?

b What is the new teff or for this drug if the dose were increased by a factor of four to 2000 mg?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 8.69 hr

WHAT THE STEM GIVES

V_D	approximately 15 L	divides the dose: C_0 = 33.33 mg/L
t½	approximately 5 hours	k = 0.1386 hr^{-1}
Ceff	10 mcg/mL	equals 10 mg/L, the MEC
D_0	500-mg	the numerator of C_0

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{0.693}{5\ hr} = 0.1386\ hr^{-1};\ Ceff = 10\ mcg/mL = 10\ mg/L$$

Her key uses the k form of the duration line, so the 5-hour half-life is turned into k first. 1 mcg/mL equals 1 mg/L, so the MEC is 10 mg/L.

ALGEBRA

$$C_0 = \frac{500\ mg}{15\ L} = 33.33\ mg/L;\ teff = \frac{\ln(33.33 \div 10)}{0.1386\ hr^{-1}} = \frac{\ln 3.333}{0.1386} = \frac{1.204}{0.1386\ hr^{-1}}$$

C_0 is the dose spread through V_D. The level only has to fall from 33.33 to 10 mg/L, a 3.33-fold fall, before the effect ends.

ROUND

$$teff = 8.69\ hr$$

1.204 ÷ 0.1386 hr^{-1} = 8.69 hr, her boxed value: between one and two half-lives of 5 hr, because 3.33 lies between 2 and 4.

> **DOES THE NUMBER MAKE SENSE?**
> C_0 ÷ Ceff = 3.333 is between 2 and 4, so teff is between one and two half-lives of 5 hr: 8.69 hr.

b

Her key

Answer: 18.69 hr

WHAT THE STEM GIVES

D_0	2000 mg	four times 500 mg, so C_0 is four times larger
V_D	approximately 15 L	unchanged by the dose
t½	approximately 5 hours	k = 0.1386 hr^{-1}
Ceff	10 mcg/mL	equals 10 mg/L

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = \frac{2000\ \text{mg}}{15\ \text{L}} = 133.3\ \text{mg/L};\ k = \frac{0.693}{5\ \text{hr}} = 0.1386\ \text{hr}^{-1};\ \text{Ceff} = 10\ \text{mg/L}$$

Four times the dose gives four times C_0, because V_D does not change with the dose; k comes from the 5-hour half-life and 10 mcg/mL equals 10 mg/L.

ALGEBRA

$$\text{teff} = \frac{\ln(133.3 \div 10)}{0.1386\ \text{hr}^{-1}} = \frac{\ln 13.33}{0.1386} = \frac{2.590}{0.1386\ \text{hr}^{-1}} = 18.69\ \text{hr}$$

The same duration line with the new C_0. Four times the ratio adds ln 4 = 1.386 to the logarithm, which is two half-lives once divided by k.

ROUND

teff = 18.69 hr; change = 18.69 − 8.69 = 10 hr = 2 t½

Her key: the change is 10 hr, two half-lives of 5 hr. A factor of four is two doublings of the dose, and each doubling adds one half-life.

DOES THE NUMBER MAKE SENSE?

Four times the dose adds two half-lives, 10 hr, to the 8.69 hr of part a: 18.69 hr, not four times 8.69.